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Overidentified IV's:

$$E[ze] = 0$$

$$E[z(Y - x^T \beta)] = 0, \quad l > k$$

$l \times 1$

$$\hat{\beta} = \underset{b}{\operatorname{argmin}} (\underline{z}^T \underline{Y} - \underline{z}^T \underline{x} b)^T \hat{W} (\underline{z}^T \underline{Y} - \underline{z}^T \underline{x} b)$$

: expand, derivative

$$\hat{\beta} = (\underline{x}^T \underline{z} \hat{W} \underline{z}^T \underline{x})^{-1} \underline{x}^T \underline{z} \hat{W} \underline{z}^T \underline{Y}$$

Picking $\hat{W}$:

$l \times l$ weighting matrix

When we pick: how much importance on each violation of moments

i.e. $\hat{W} = I$, treat all violations equally.

put more weight on moments by variance (diagonal), covariance (non-diagonal)

↳ more weight on low var, cov.

use $\hat{W} = (\text{variance matrix} := \underline{\Omega})^{-1}$

$$\underline{\Omega} = \operatorname{var}(ze) = E[ze ze^T]$$

$$\hat{W} \xrightarrow{P} W; \quad W = \underline{\Omega}^{-1}$$

$$\hat{\underline{W}} = \hat{\underline{\Omega}}^{-1} = \left(\frac{1}{n} \sum_{i=1}^n \underline{z}_i \underline{z}_i^T \hat{e}_i^2 \right)^{-1} \quad \text{optimal GMM (look at 13.8) for efficiency}$$

In PS6:

$$\sqrt{n}(\hat{\beta}_g - \beta) \xrightarrow{d} N(0, \underline{V})$$

where $\underline{V} =$

$$([E[XZ^T] \underline{W} E[ZX^T]])^{-1} E[XZ^T] \underline{W} \underline{\Omega} \underline{W} E[ZX^T] (E[XZ^T] \underline{W} E[ZX^T])^{-1}$$

↳ to estimate $\Rightarrow$ expectation $\rightarrow \frac{1}{n} \sum_{i=1}^N$; $\underline{W} \rightarrow \hat{\underline{W}}$

$$\hat{\beta}_g^0 = (\underline{X}^T \underline{Z} \hat{\underline{\Omega}}^{-1} \underline{Z}^T \underline{X})^{-1} \underline{X}^T \underline{Z} \hat{\underline{\Omega}}^{-1} \underline{Z}^T \underline{Y}$$

$$\sqrt{n}(\hat{\beta}_g^0 - \beta) \xrightarrow{d} N(0, \underline{V}_0)$$

$$\underline{V}_0 = (E[XZ^T] \underline{\Omega}^{-1} E[ZX^T])^{-1}$$

to get $\hat{\underline{\Omega}}^{-1}$, need estimate $\underline{\Omega}$. need residuals, choose $\hat{\underline{W}}$ to identify like I to get $\hat{\beta}$, get $\hat{e}_i^2$, get $\hat{\underline{\Omega}}^{-1}$.

2 step GMM

$\underline{W} = (\underline{Z}^T \underline{Z})^{-1}$ is another choice, yielding:

$$\hat{\beta} = (\underline{X}^T \underline{Z} (\underline{Z}^T \underline{Z})^{-1} \underline{Z}^T \underline{X})^{-1} \underline{X}^T \underline{Z} (\underline{Z}^T \underline{Z})^{-1} \underline{Z}^T \underline{Y}$$

↳ 2sls

Overidentification Tests:

Ex: $k=1, l=2$

(1) $0 = E[z_1 e] = E[z_1 (Y - x_1 \beta)]$

(2) $0 = E[z_2 e] = E[z_2 (Y - x_1 \beta)]$

recover $\hat{\beta}$ w/ (1), use to est. (2)

$$J(b) = n \left(\frac{1}{n} \underline{z}^T \underline{Y} - \frac{1}{n} \underline{z}^T \underline{X} b \right)^T \hat{\Omega} \left(\frac{1}{n} \underline{z}^T \underline{Y} - \frac{1}{n} \underline{z}^T \underline{X} b \right)$$

$H_0: E[ze] = 0, H_1: E[ze] \neq 0$

under $H_1: J(b) \rightarrow \infty$

J -stat H_0 : act like a distribution $\rightarrow (*)$

$J(\hat{\beta}_g) = \frac{\left(\sqrt{n} \hat{\Omega}^{-\frac{1}{2}} \left(\frac{1}{n} \underline{z}^T \underline{Y} - \frac{1}{n} \underline{z}^T \underline{X} \hat{\beta}_g \right) \right)^T}{\left(\sqrt{n} \hat{\Omega}^{-\frac{1}{2}} \left(\frac{1}{n} \underline{z}^T \underline{Y} - \frac{1}{n} \underline{z}^T \underline{X} \hat{\beta}_g \right) \right)}$
 $\hookrightarrow$ converge to χ^2

look at term 1:

$$\begin{aligned} & \sqrt{n} \left(\hat{\Omega}^{-\frac{1}{2}} \left(\frac{1}{n} \underline{z}^T \underline{Y} - \frac{1}{n} \underline{z}^T \underline{X} \hat{\beta}_g \right) \right) \\ &= \sqrt{n} \left(\hat{\Omega}^{-\frac{1}{2}} \left(\frac{1}{n} \underline{z}^T \underline{e} - \frac{1}{n} \underline{z}^T \underline{X} (\hat{\beta}_g - \beta) \right) \right) \\ &= \sqrt{n} \left(\hat{\Omega}^{-\frac{1}{2}} \left(\frac{1}{n} \underline{z}^T \underline{e} - \frac{1}{n} \underline{z}^T \underline{X} (\underline{X}^T \underline{Z} \hat{\Omega}^{-1} \underline{Z}^T \underline{X})^{-1} \underline{X}^T \underline{Z} \hat{\Omega}^{-1} \underline{z}^T \underline{e} \right) \right) \\ & \quad \hookrightarrow \text{factor out } \underline{z}^T \underline{e} \\ &= \left(\underline{I} - \frac{1}{n} \underline{z}^T \underline{X} \left(\frac{1}{n} \underline{X}^T \underline{Z} \hat{\Omega}^{-1} \underline{Z}^T \underline{X} \right)^{-1} \frac{1}{n} \underline{X}^T \underline{Z} \hat{\Omega}^{-\frac{1}{2}} \right) \hat{\Omega}^{-\frac{1}{2}} \frac{1}{\sqrt{n}} \underline{z}^T \underline{e} \\ & \quad - \text{Let } \underline{R} = \hat{\Omega}^{-\frac{1}{2}} E[\underline{z} \underline{z}^T] \\ & \quad \hat{\Omega}^{-\frac{1}{2}} \frac{1}{n} \underline{z}^T \underline{X} \xrightarrow{p} \underline{R}; \quad \frac{1}{\sqrt{n}} \underline{z}^T \underline{e} \xrightarrow{d} N(0, \underline{\Omega}), \quad \hat{\Omega}^{-\frac{1}{2}} \frac{1}{\sqrt{n}} \underline{z}^T \underline{e} \xrightarrow{d} N(0, \underline{I}) \end{aligned}$$

$\star \xrightarrow{d} (\underline{I} - \underline{R}(\underline{R}^T \underline{R}) \underline{R}^T) u$ s.t. $u \sim N(0, \underline{I})$, under H_0

$$J(\hat{\beta}_g^0) \xrightarrow{d} u^T (\underline{I} - R(R^T R)^{-1} R^T) (\underline{I} - R(R^T R)^{-1} R^T) u$$

→ idemp. & symm like annihilator

$$= u^T (\underline{I} - R(R^T R)^{-1} R^T) u \sim \chi_{\ell-k}^2$$

$$\text{tr}(\underline{I} - R(R^T R)^{-1} R^T) = \ell - k$$

J-stat behaves like draw from χ^2 , look up critical values, if really big moment conditions specified incorrectly.